

PHYS 427: Lecture # 7

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Lecture Span

1. Helmholtz Free Energy

Recap

At high temperatures, each quadratic DOF gives $k_B T/2$ to the energy.

Theorem 0.1. *Equipartition theorem.*

$$\epsilon = a_1 q_1^2 + a_2 q_2^2 + \dots + a_n q_n^2 \quad (1)$$

Here, q_i are continuous variables varying from $-\infty$ to ∞ . For example,

$$\epsilon = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2 \quad (2)$$

In the high temperature limit, you turn the partition function into integrals. Then

$$Z(T) \approx A \int dq_1 \dots dq_n e^{-\epsilon_i(q_1, q_2, \dots, q_n)\beta} \quad (3)$$

Then,

$$p(\epsilon_i) = e^{-\epsilon_i/k_B T} / Z(T) = A \frac{dq_1 \dots dq_n e^{-\beta \epsilon(q_i)}}{A \int [\dots]} \quad (4)$$

Then,

$$\langle \epsilon_{q_1} \rangle = \langle a_1 q_1^2 \rangle = \frac{\int dq_1 \dots a_1 q_1^2 e^{-\beta \epsilon(q_i)}}{Z(T)} \quad (5)$$

The integrals upstairs and downstairs cancel. Then, this fraction is

$$= \frac{\int_{-\infty}^{\infty} dq_1 a_1 q_1^2 \exp(\dots a_1 q_1)}{\int \exp(\dots a_1 q_1)} \quad (6)$$

$$= -\frac{\partial}{\partial \beta} \ln \left[\int dq_1 \exp(\dots a_1 q_1^2) \right] \quad (7)$$

$$= -\partial_\beta \ln \left(\beta^{-1/2} \right) \quad (8)$$

$$= \frac{1}{2\beta} = \frac{1}{2} k_B T \quad (9)$$

This is simply the equipartition theorem (only valid in the high temperature limit!).

Example: Maxwell velocity distribution

Probability for an atom in a gas to have speed in the range $[v, v + dv]$. For this, use spherical coordinates. Here,

$$\langle v \rangle = \sqrt{\frac{8k_B T}{\pi m}} \quad (10)$$

$$\langle v^2 \rangle = \frac{3k_B T}{m} \quad (11)$$

Helmholtz Free Energy

To recap, we have had the microcanonical ensemble (U is fixed, and volume of the system is fixed), then the entropy of this system increases until maximum.

The canonical ensemble is if the system is coupled to a reservoir R that is much larger than the system. Both have a fixed temperature of $T = T_S = T_R$, fixed by the reservoir. Also, volume of the system is fixed.

The total energy $U_R + U_S$ is fixed. But both can individually vary. Therefore, the entropy of the system is max because $R + S$ is closed.

Is there a quantity of the system alone that is at an extremum? This is the Helmholtz free energy! Here, the number of particles N is fixed.

Start with $S(U, V)$, then

$$dS = \partial_U S dU + \partial_V S dV \quad (12)$$

$$= \frac{1}{T} dU + \frac{p}{T} dV \quad (13)$$

$$= \frac{1}{T} dU \quad (14)$$

Assume that volume is fixed, or $dV = 0$. Here, $dS_{R+S} = 0 = dS_R + dS_S$. Then

$$= \frac{dU_R}{T} + dS_S \quad (15)$$

Moreover, $dU_R = -dU_S$. Then,

$$dS = -\frac{dU_S}{T} + dS_S = -\frac{1}{T} d(U_S - TS_S) \quad (16)$$

Thus,

$$-\frac{1}{T} dF = -\frac{1}{T} d(U_S - TS_S) \quad (17)$$

Here, F is the Helmholtz free energy. Therefore, if $R + S$ is at equilibrium, then F is at a minimum.

The thermodynamic identity is below:

$$dS = \frac{1}{T} dU + \frac{p}{T} dV \iff dU = TdS - pdV \quad (18)$$

This is also the first law of thermodynamics. TdS is the heat and pdV is the work done on the system. For Helmholtz free energy,

$$dF = dU - TdS - SdT \quad (19)$$

$$= TdS - pdV - TdS - SdT \quad (20)$$

$$= -pdV - SdT \quad (21)$$

$$= \frac{\partial F}{\partial V} dV + \frac{\partial F}{\partial T} dT \quad (22)$$

Therefore, $F = F(T, V)$. Canonical ensemble is fixed by the temperature and the volume. Therefore,

$$p = -\frac{\partial F}{\partial V}, \quad S = -\frac{\partial F}{\partial T} \quad (23)$$

Is there a relationship between F and Z ? Consider a microcanonical ensemble, then

$$S_S = k_B \ln(\Omega) \quad (24)$$

In the canonical ensemble, Z is like Ω such that:

$$F = -k_B T \ln(Z) \quad (25)$$

A quick sketch of this formula:

$$F = U - TS \quad (26)$$

$$\iff S = \frac{U - F}{T} \quad (27)$$

$$\frac{\partial F}{\partial T} = \frac{F - U}{T} \quad (28)$$

Here, equation (25) solves the above diff eq. Here,

$$p = k_B T \partial_V \ln(Z) \quad (29)$$

$$S = \partial_T (k_B T \ln(Z)) \quad (30)$$

For an ideal gas, $Z = (n_Q V)^N / N!$.

Quantum density:

$$n_Q = \left(\frac{mk_B T}{2\pi\hbar^2} \right)^{3/2} \quad (31)$$

Then we use the above equations to derive:

$$p = \frac{Nk_B T}{V} \iff pV = Nk_B T \quad (32)$$

Moreover, using Stirling's approximation:

$$S = -F/T + U/T \quad (33)$$

$$= Nk_B \ln(n_Q V) - Nk_B \ln N + Nk_B + k_B T \partial_T \ln(T^{-3/2}) \quad (34)$$

$$= Nk_B \left[\ln \left(\frac{n_Q V}{N} \right) + 5/2 \right] \quad (35)$$

This is called the Sackur-Tetrode equation.

Example

System with two energy levels. Calculate S and $T \rightarrow \infty, T \rightarrow 0$. The partition function $e^0 + e^{-\beta\Delta} = 1 + e^{-\beta\Delta}$. Then,

$$S = \partial_T (k_B T \ln(Z)) = k_B T \ln \left(1 + e^{-\Delta/k_B T} \right) + \frac{\Delta}{T} \frac{\exp(-\Delta/k_B T)}{1 + \exp(-\Delta/k_B T)} \quad (36)$$

As $T \rightarrow 0$, $S \rightarrow 0$. As $T \rightarrow \infty$, then $S \rightarrow k_B \ln(2)$.